

FROM CIFAR CLASSIFIERS TO CLIP ENCODERS

Invariance is not robustness

What actually predicts a model's adversarial robustness

Anass Al Ammiri

Technical University of Munich

A model that ignores small shifts of its input is called invariant. A model that does not flip under a tiny worst-case change is called robust. We show these are different, measure what tells them apart, and correct a common mistake.

The puzzle

Does making a vision model more shift-invariant help or hurt its adversarial robustness?

One literature says: it helps.

Anti-aliasing and equivariant designs report better robustness. Smoother, more invariant models look safer.

Another says: it hurts.

Perfect shift-invariance can shrink the margin between classes, which should make attacks easier.

Both can't be right as stated. Which is it, and what is the real signal?

Takeaway. We resolve this with one simple quantity, and find that the field's usual invariance metric is the wrong thing to look at.

Two different things, often confused

Invariance

Ignoring changes that should not matter:
a small shift, a crop, a reworted prompt.



A shift is a change we *want* the model to ignore.

Adversarial robustness

Not being flipped by a small *worst-case*
change chosen by an attacker.

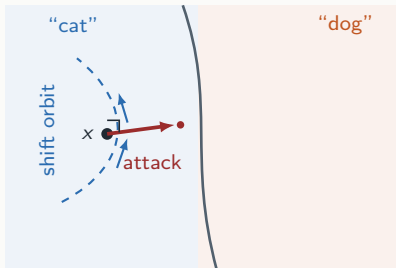


An attack is a change the model *must not* be
fooled by.

They sound related, and the field often uses the first as a proxy for the second. Our first finding is that this proxy does not work.

The picture: two different directions

Invariance moves *along* the boundary; an attack moves *across* it



- **Invariance** is being *flat along the shift orbit*: the dashed path stays inside the "cat" region, so the label never changes.
- **Robustness** is the *margin to the boundary*: the red arrow is the smallest change that crosses into "dog".
- These are **different directions**. A long, flat orbit (very invariant) can still sit a hair from the boundary (fragile).
- Invariance is a *projection*; robustness is a *distance*. One says little about the other.

Takeaway. Invariance and robustness live along near-orthogonal directions. This is the geometric reason a good invariance score does not buy you robustness.

The measuring stick: a margin-to-sensitivity ratio

η/L , computed on clean images, no attack needed

$$\eta/L = \frac{\text{margin}}{\text{sensitivity}} = \frac{\text{how confident the model is}}{\text{how fast that confidence changes if you nudge the input}}$$

- The **margin** is the gap between the top class and the runner-up.
- The **sensitivity** is the size of the input gradient of that margin, measured in the way the attacker moves (the norm matched to the threat).
- A large ratio means: far from the decision boundary, and slow to get there. That is exactly what robustness should be.

Takeaway. η/L is a first-order estimate of how big a perturbation it takes to flip the label. It needs no attack and is not changed by rescaling the model's outputs.

Finding 1: the invariance metric is saturated

Shift-consistency looks the same for every modern model, robust or not

	shift-consistency	spread across models
16 frozen CLIP encoders (robust and non-robust alike)	0.96 – 0.99	almost none

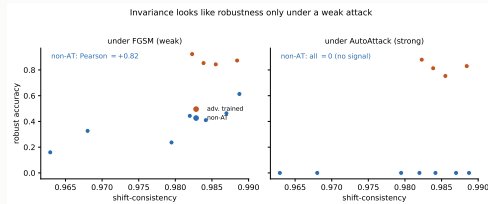
- Every modern encoder is already near-perfectly shift-consistent, so the metric cannot tell them apart on *any* kind of robustness.
- Once you account for clean accuracy, it does not predict robustness to common corruptions either.
- The same squeeze appears on strong CNN backbones: consistency stops discriminating.

Takeaway. Shift-consistency is treated as a robustness signal, but it is saturated. It has no room left to rank models. This is why the “invariance predicts robustness” story keeps giving mixed answers.

Finding 2: “invariance helps” is a weak-attack artifact

The effect shows up under a cheap attack and vanishes under a strong one

- Under a single-step FGSM attack, more invariant models look more robust: the “invariance helps” ordering appears (left panel, a real trend).
- Under AutoAttack, every non-robust encoder drops to **zero**. There is no robustness left to order, and the trend disappears (right panel).
- A weak attack simply overstates robustness, the classic gradient-masking signature.



Same encoders, same budget: shift-consistency looks like it predicts robustness under FGSM, and predicts nothing under AutoAttack.

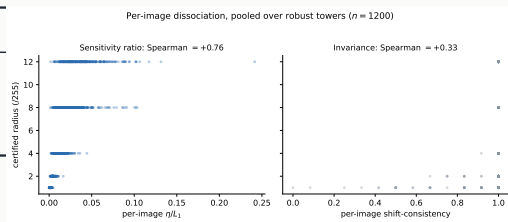
Takeaway. Recent claims that architectural invariance gives robustness without adversarial training are measured with weak attacks. They do not survive a strong one.

Finding 3: the margin ratio predicts robustness

η/L orders robustness where invariance does not, across three settings

setting	η/L	consistency
CIFAR-10 classifiers	+0.99	uninformative
ResNet, ImageNet-100	+0.90	anti-corr.
CLIP, per image	+0.74–0.96	≈ 0

- The same law from small CNNs to a ResNet to frozen foundation-model encoders.
- Inside one robust encoder, image by image, η/L tracks how large an attack each image needs; shift-consistency does not.

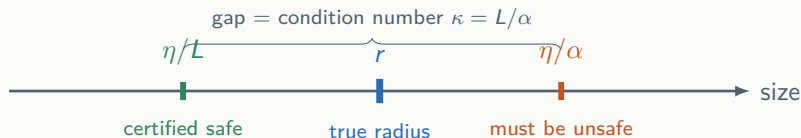


Per image, within a robust CLIP encoder: η/L (left) orders the attack size, shift-consistency (right) does not.

Takeaway. The right question is not “how invariant is the model” but “how large is the margin relative to the sensitivity”. That is η/L .

Why it works: a two-sided bracket

The ratio is not just a heuristic; it brackets the true robust radius



- The **lower bound** is the standard certificate: the model cannot be flipped by anything smaller than η/L .
- The **upper bound** is the matching side: with α a slope along the path to the boundary, no perturbation larger than η/α can be safe.
- The gap between the two is a condition number $\kappa = L/\alpha$. When it is close to one, η/L is not just a bound, it is the answer.

Takeaway. η/L is the small-curvature limit of the true radius. This tells us exactly when it is tight, and points to what it misses.

Finding 4: detecting robustness is not the same as ranking it

η/L separates robust from fragile, but cannot order the robust ones

- Across all encoders, η/L cleanly **separates** the robust ones from the fragile ones.
- But *among the already-robust* encoders, η/L no longer orders them. Two encoders with the same η/L can differ in robustness.
- Something beyond the margin and the size of the gradient is at play.

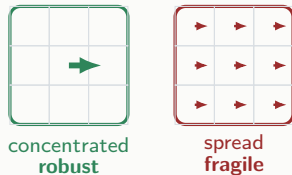
So what ranks the robust models?

Takeaway. η/L uses the *size* of the gradient. To rank robust models we need its *shape*.

Finding 4: the shape of the gradient ranks robust models

A concentrated gradient resists an all-pixel attack better than a spread one

- An ℓ_∞ attack pushes on *every* pixel at once.
- If the margin's gradient is **concentrated** on a few pixels, most of that budget is wasted, and the model holds.
- If it is **spread** over many pixels, the attack engages all of them, and the model gives way.
- Measured by the gradient's **anisotropy** (how spread it is), this orders the robust encoders where η/L could not.



An all-pixel attack wastes its budget on a concentrated gradient, but exploits every arrow of a spread one.

We measured this shape by the gradient's **anisotropy** $A = \|\nabla M\|_1 / \|\nabla M\|_2$: near 1 when concentrated,

up to $\sqrt{d}$ when spread. It is the same η/L ingredients, read as a *ratio of norms* instead of a size.

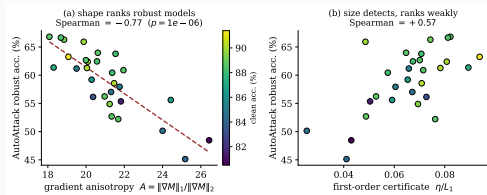
Finding 4: why shape matters, and that it holds on real models

An effective-dimension deficit, not curvature, and it replicates on 30 CIFAR models

- η/L_1 is only the start. A real attack also harvests the gradient's variations across pixels, collecting *more* when the gradient is spread over more coordinates.
- This shrinks the true radius below η/L_1 by a factor growing with the spread:

$$r \approx \frac{\eta/L_1}{1 + CA}$$

- A **transverse** effect, *not* curvature along the attack (on-direction curvature does **not** rank robustness). A matched-margin synthetic model reproduces the ordering exactly (rank corr. -1.0).



30 RobustBench CIFAR-10 models. (a) the shape A ranks robustness (rank corr. -0.77 , survives a clean-accuracy control); (b) the size η/L_1 detects it but ranks weakly.

Takeaway. The shape law has a mechanism and replicates on 30 independent real robust models.

Finding 5: adversarial training flips a hidden relationship

A fingerprint of what robust training actually changes

- In an ordinary model, the images with the largest margin tend to have the *smallest* gradients: they sit in flatter regions, so margin and robustness reinforce.
- After adversarial training, this reverses: high-margin images now have the *largest* gradients, so the model buys margin partly by raising sensitivity.
- The per-image relationship between margin and sensitivity flips sign, consistently across dozens of models.

Takeaway. Robust training does not just raise the margin. It re-couples margin and sensitivity, and η/L is the quantity that reads this change.

What it all means

1. **Stop using shift-consistency as a robustness signal.** It is saturated on modern models and does not order adversarial robustness.
2. **Distrust “invariance helps” claims tested with weak attacks.** They vanish under AutoAttack.
3. **Use the margin-to-sensitivity ratio η/L .** It predicts robustness without running an attack, from CNNs to frozen CLIP encoders.
4. **To compare strong models, look at the gradient’s shape,** not only its size. Concentrated sensitivity is more robust.

One idea, the margin relative to the sensitivity, unifies a split literature and turns a vague intuition about invariance into something you can measure and predict.

How to present this

The whole story in three beats

The three-beat story

- Beat 1. The confusion.** People treat invariance and robustness as the same thing and get contradictory results. They are not the same thing.
- Beat 2. The fix.** A simple ratio, margin over sensitivity, predicts robustness without any attack, and the field's invariance metric does not, because it is saturated.
- Beat 3. The refinement.** That ratio tells robust from fragile; to rank the robust ones you also need the shape of the model's sensitivity, and the apparent “invariance helps” effect is just a weak-attack mirage.

Takeaway. If you remember one line: *invariance is not robustness; the margin relative to the sensitivity is.*

Two analogies and the questions you will get

- **Analogy 1.** A locked door is not a strong door. Ignoring a gentle push (invariance) is not the same as withstanding a crowbar (an attack).
- **Analogy 2.** Two people equally far from a cliff edge: the one who moves slowly toward it is safer. That “distance over speed” is η/L .

Q: Isn't the ratio just re-describing the radius it predicts? Partly, and we say so; the surprise is that the invariance metric fails where the ratio succeeds.

Q: Does it hold on big models? Yes, on frozen CLIP encoders as well as CNNs.

Q: Can you optimize it directly? No, it is a diagnostic, not a training target; forcing it up collapses the margin.